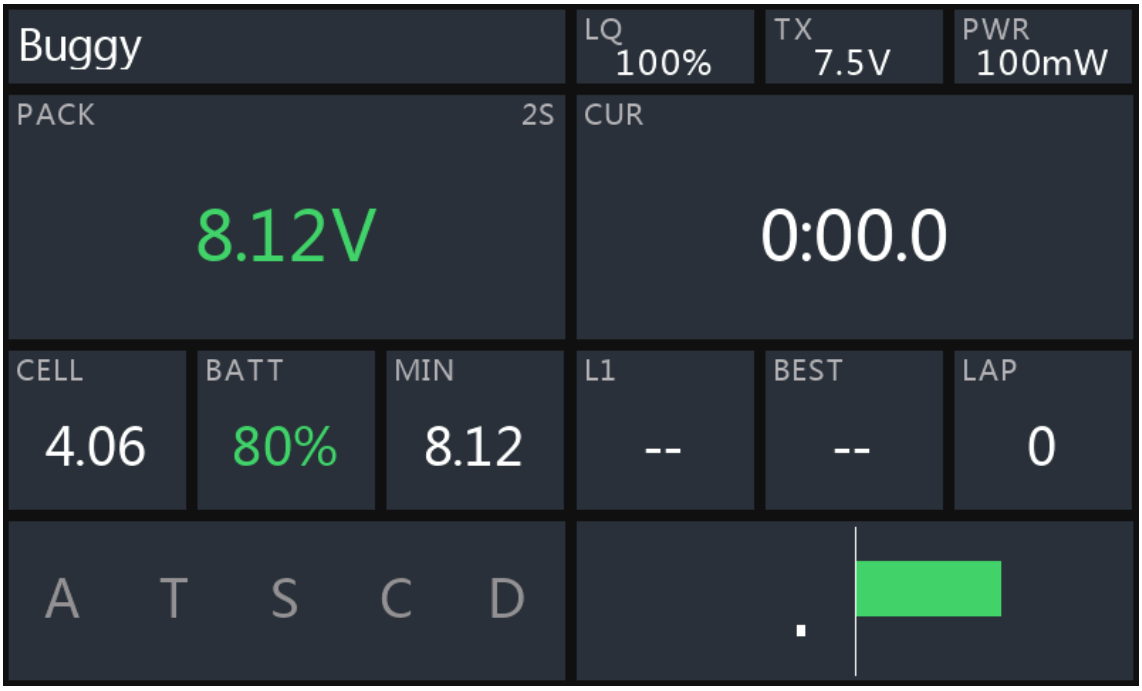
Trackside

Race dashboard and drive aids
for the FrSky Velox VX10



The Trackside dashboard on a VX10

Setup guide | Version 0.1.0 | 26 October 2026

Before you start

This guide sets up Trackside on a car model, from a fresh model through to the dashboard. It takes about ten minutes. Every screenshot is from the VX10.

You need	
Radio	FrSky Velox VX10 with Ethos 26.2 or later
Software	Trackside 0.1.0 (the **trackside** folder)
Model	A car model. Section 2 creates one with the Ethos wizard.

How Trackside connects to your car

Trackside adds two items to your radio:

Part	What it is	What it does
Throttle Aid and **Steer Aid**	Lua sources	Each outputs a multiplier for one channel. Your mixes keep the normal stick line and add one **Multiply** line with the aid after it.
Trackside	Widget	The race dashboard. Its Configure page holds all Trackside settings.

Because the stick line stays in place, the car is always driven by your trigger and wheel. The aids only scale them. If Trackside ever stops running, the channel falls back to the plain stick.

About this version. The aids are set up and connected, but in version 0.1.0 they pass the sticks through unchanged. Setting them up now means nothing else needs to change when the aids arrive in an update.

Safety. Make mixer changes on the bench with the car's motor disconnected or the radio's RF output off. Check failsafe still works afterwards.

1. Install Trackside

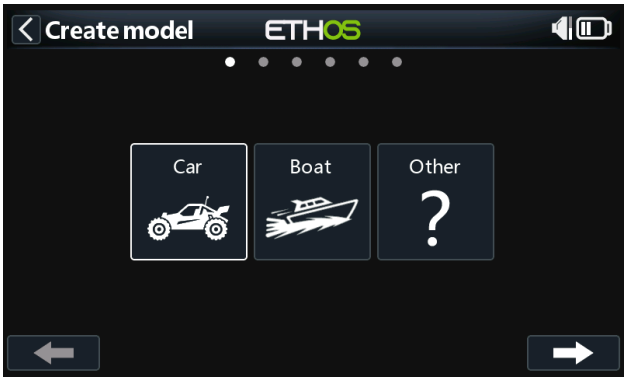
1. Connect the VX10 to your computer by USB and choose **USB storage** on the radio.
2. Copy the **trackside** folder into the radio's **SCRIPTS** folder, so you have **SCRIPTS/trackside/main.lua**.
3. Eject the radio drive safely and unplug it. The radio loads Trackside on its next start.

> Your settings are saved in **SCRIPTS/trackside.user**, a separate folder. Updating Trackside replaces only **SCRIPTS/trackside**, so your settings are kept.

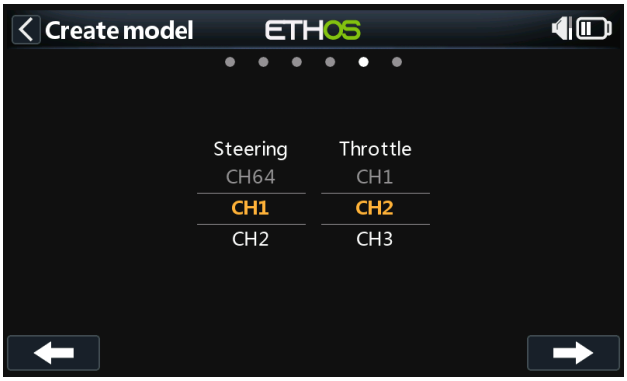
2. Create a car model

Skip this if you already have a car model with Steering on CH1 and Throttle on CH2. Otherwise create one with the Ethos wizard: **Model > Model select > +**.

1. Choose **Car**.
2. Step through the engine, steering and brake pages. The defaults are fine for most cars.
3. On the channel page, keep **Steering = CH1** and **Throttle = CH2**. The rest of this guide uses these.
4. Give the model a name and finish the wizard.



Choose Car



Steering CH1, Throttle CH2

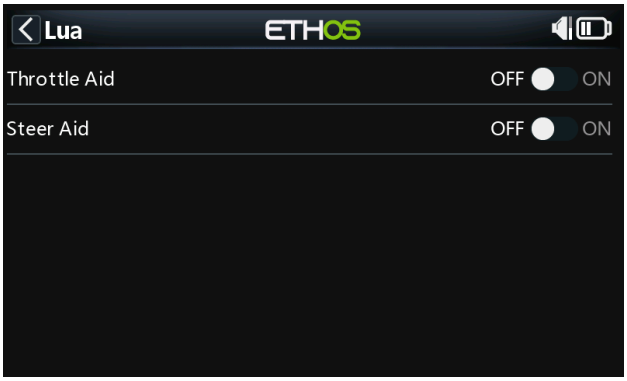
3. Turn on Throttle Aid and Steer Aid

Ethos only runs a Lua source, and only lists it when you pick a mix source, after you turn it on for the model. Do this once per model.

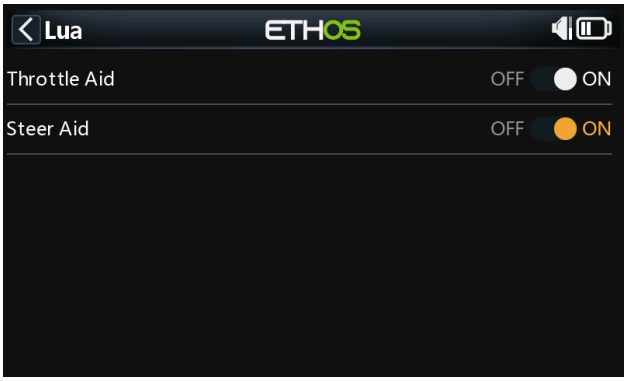
1. Open **Model** and swipe to the second page.
2. Tap **Lua**.
3. Turn **Throttle Aid** and **Steer Aid** ON.



Model menu, page 2: the Lua page



Both start OFF

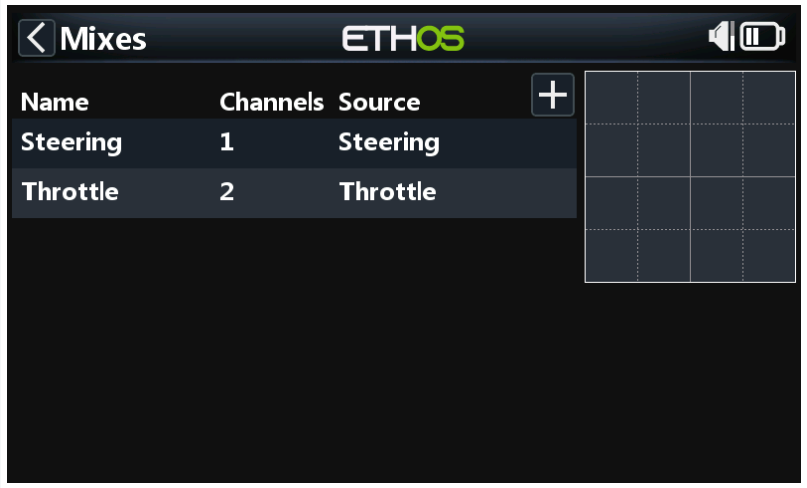


Turn both ON

4. Add the aid mixes

The car wizard made two mixes: **Steering** (CH1) and **Throttle** (CH2). **Leave them exactly as they are.** You add one more mix after each.

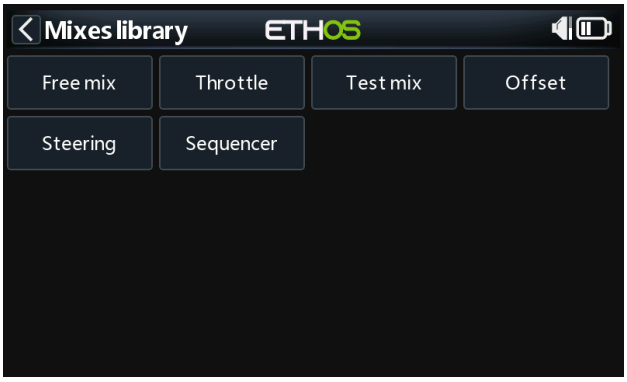
> **Don't** change the source of the existing Steering or Throttle mix to an aid. The channel would then follow the aid's fixed value instead of your stick, and nothing would move.



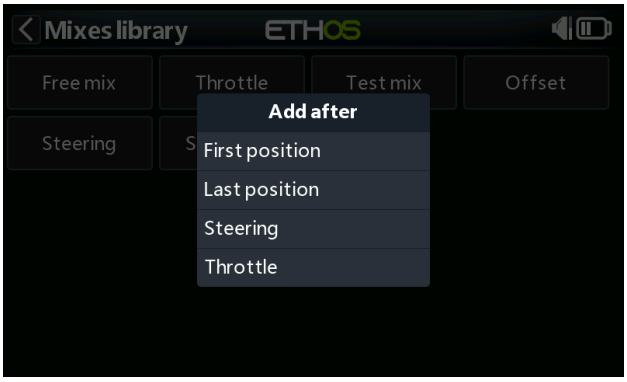
Model > Mixes: the two mixes from the wizard

4a. Throttle

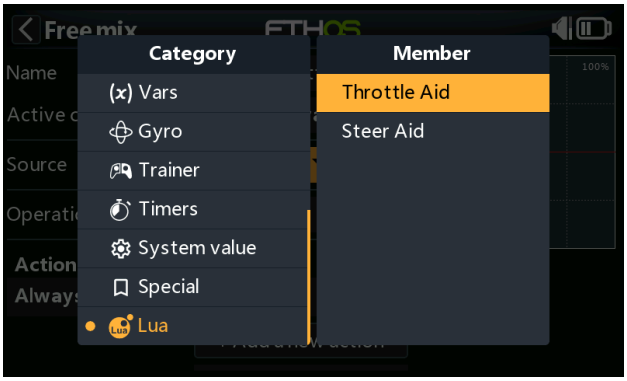
1. In **Mixes**, tap **+**, choose **Free mix**, then **Add after: Throttle**.
2. Optional: set **Name** to *Throttle Aid* so it's easy to spot in the list.
3. Tap **Source**, scroll to the **Lua** category and pick **Throttle Aid**.
4. Set **Operation** to **Multiply**.
5. Scroll down and set **Output** to **CH2 (Throttle)**. A new Free mix starts with Output = **---** and does nothing until you set it.



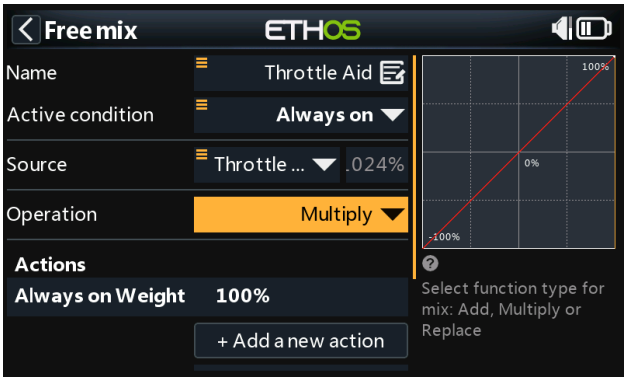
+ opens the mix library: choose Free mix



Add after: Throttle



Source: Lua > Throttle Aid



Operation: Multiply

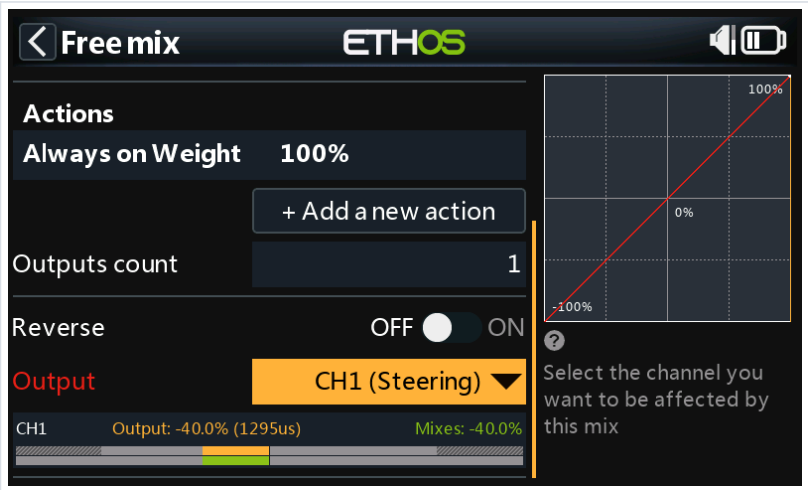


Output: CH2. The bar shows the result: throttle 60% gives CH2 60%.

> The grey box next to Source shows about **1024%**. That's Ethos showing the aid on its internal scale, and it's expected. What counts is the channel output.

4b. Steering

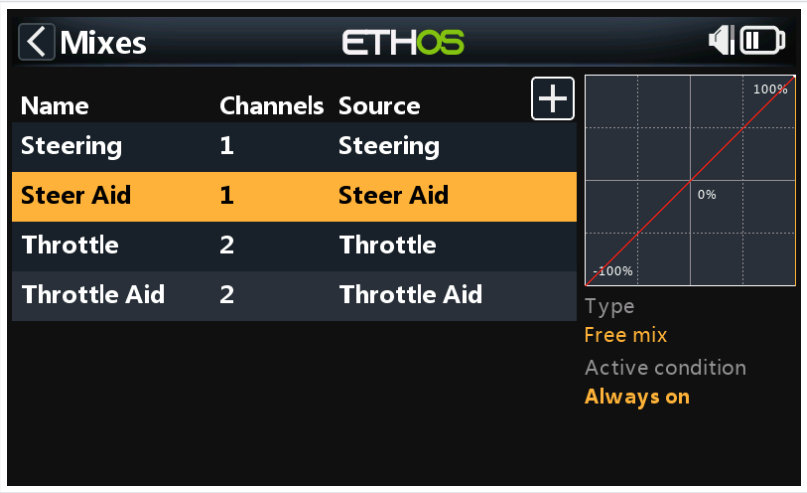
Repeat the same steps with these changes: **Add after: Steering**, source **Lua > Steer Aid**, Operation **Multiply**, Output **CH1 (Steering)**.



Steer Aid on CH1: wheel at -40% gives CH1 -40%

4c. Check the order

Each stick mix should be followed by its aid. Use exactly one aid line per channel: a second Multiply line with the same aid applies it twice.



Finished: Steering then Steer Aid (CH1), Throttle then Throttle Aid (CH2)

Channel	Mix	Source	Operation	Output
CH1	Steering	Analogs > Steering (the wheel)	Add	CH1
CH1	Steer Aid	Lua > Steer Aid	Multiply	CH1
CH2	Throttle	Analogs > Throttle (the trigger)	Add	CH2
CH2	Throttle Aid	Lua > Throttle Aid	Multiply	CH2

If you need to reverse a channel, do it on the stick mix or in Channels, never on the aid line.

5. Check the channels

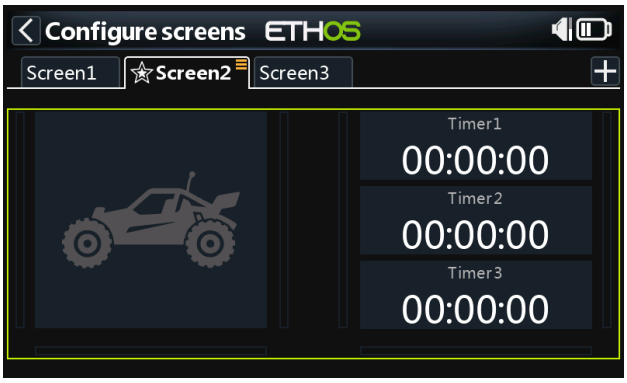
Open **Model > Channels**, then move the wheel and pull the trigger. CH1 should follow the wheel and CH2 the trigger, the same as before you added the aids.



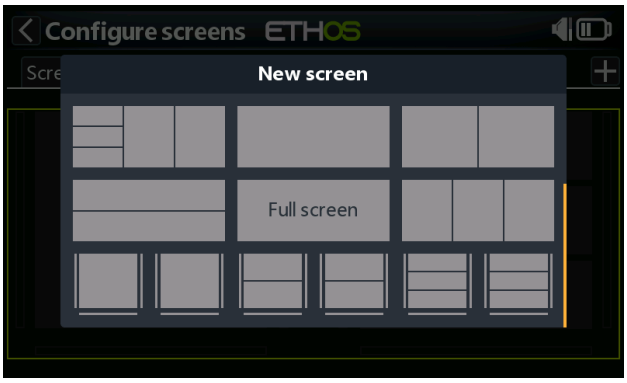
Wheel at 25% and trigger at 50%: CH1 25%, CH2 50%

6. Add the dashboard

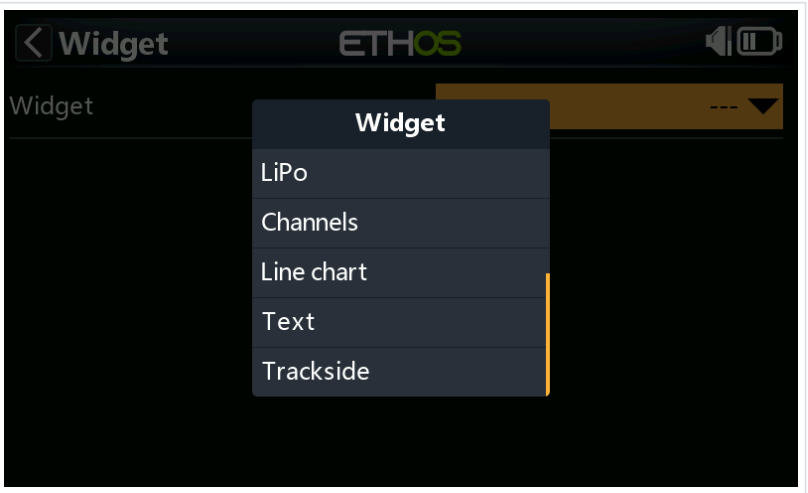
1. Tap the **Screens** icon in the bottom bar (the square with a +).
2. Tap **+** at the top right to add a screen.
3. Choose **Full screen**. Trackside also works in other layouts; in full screen it adds its own header with model name, link quality, radio battery and transmit power.
4. Tap the empty widget area, then pick **Trackside** from the widget list.
5. Go back to the home screen and swipe to the new screen.



Screens: tap + to add a screen



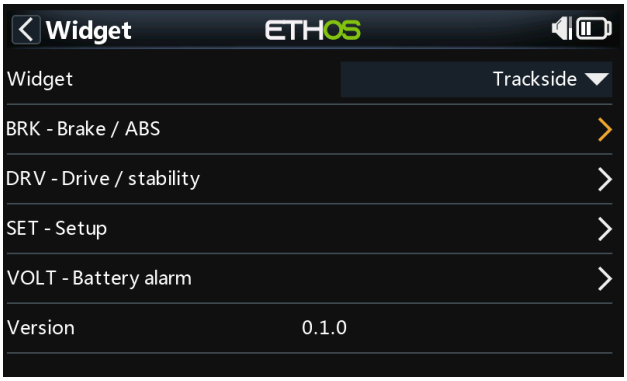
Choose Full screen



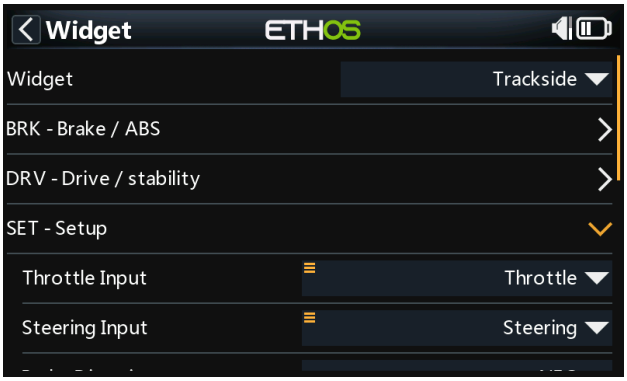
Pick Trackside from the widget list

7. Trackside settings

All settings are on the widget's own **Configure** page: in **Screens**, tap the Trackside widget. They're grouped into four panels; tap a panel to open it. Settings are saved for each model when you leave the page.



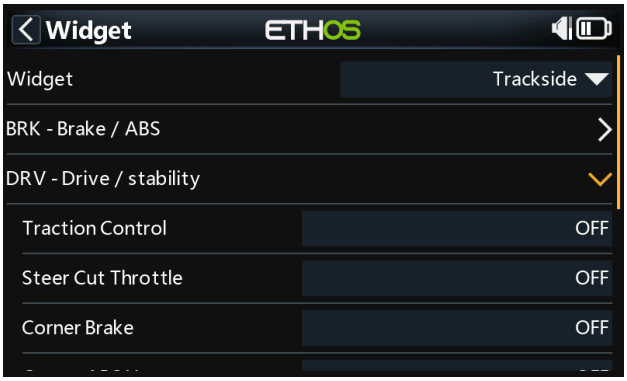
The four settings groups



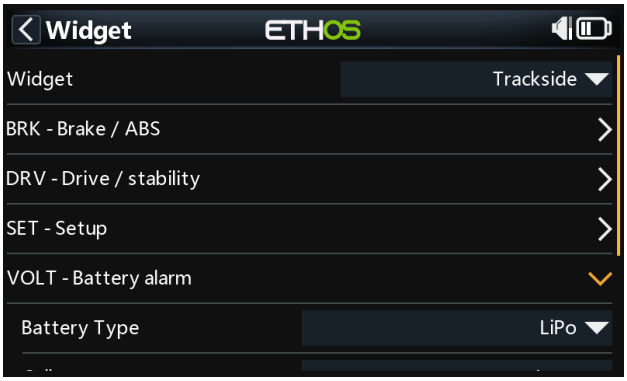
SET: inputs and directions

Check these first

Setting	Group	What to set
Throttle Input	SET	The control the aids read for throttle. Default: the VX10 trigger (Throttle).
Steering Input	SET	The control the aids read for steering. Default: the VX10 wheel (Steering).
Brake Direction	SET	Which way the trigger brakes. Pull the brake and watch the dashboard bar: it should turn red.
Battery Type, Cells	VOLT	Match your packs (LiPo or LiHV). Cells = Auto detects the cell count.
Volt Alarm, Alarm Cell Voltage	VOLT	Turn the low-voltage alarm on and choose the per-cell warning level.



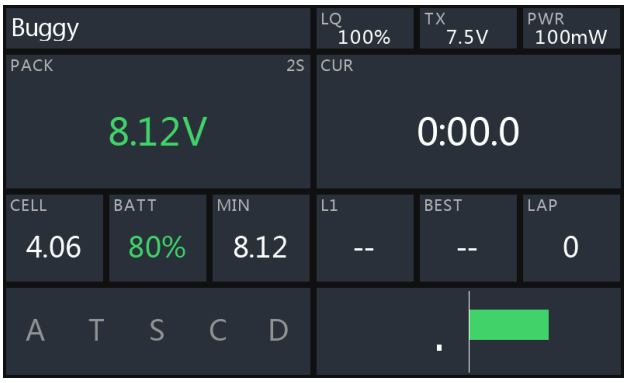
DRV: drive aids (OFF by default)



VOLT: battery alarm

The BRK (brake pulse) and DRV (drive aid) settings are all OFF or gentle by default. Leave them there for now: see *About this version* on page 2.

8. Reading the dashboard



On throttle: the bar fills green to the right



On the brake: the bar fills red

Area	Shows
Header (full screen)	Model name, link quality (LQ), radio battery (TX) and transmit power (PWR).
PACK	Car pack voltage, with the detected cell count (for example 2S) in the corner. Green above 50%, amber below 50%, red below 20%.
CELL / BATT / MIN	Voltage per cell, estimated charge left, and the lowest pack voltage this session.
CUR / L1 / BEST / LAP	Lap timing: current lap, last lap, best lap and lap count. The lap timer arrives in a later version.
A T S C D	Drive aid indicators. Each lights in its own colour while that aid is working.
Live bar	Throttle (green) or brake (red) from the centre line. The small mark below shows the wheel position.

Telemetry values show **--** until the receiver sends them.

9. Troubleshooting

Problem	Check
A channel doesn't move at all	The stick mix's source was changed to an aid. Set the Steering and Throttle mixes back to Analogs > Steering and Analogs > Throttle, and add the aids as separate Multiply lines (section 4).
Throttle or steering moves only a little	Update Trackside to the latest version. Early test builds scaled the aid output wrongly.
The aid mix has no effect	Set the Free mix's **Output** to the channel (CH2 or CH1). Also check that Operation is **Multiply**, not Add.
No Lua category when picking a source	Turn on Throttle Aid and Steer Aid in **Model > Lua** (section 3). This is needed once per model.
Trackside isn't in the widget list	Check the folder is **SCRIPTS/trackside** with **main.lua** inside, then restart the radio.
The dashboard shows -- everywhere	No telemetry yet. Values appear once the receiver is bound and sending telemetry.
The bar turns red when I accelerate	Change **Brake Direction** in the SET settings.